

Telco Customer Churn Analysis & Visualization

Project Overview

This project focuses on analyzing customer data to understand churn behavior by creating various visualizations that highlight patterns, trends, and key factors influencing customer decisions. The main objective is to explore customer characteristics such as tenure, contract type, services, and billing information to identify what leads to churn and how businesses can improve retention.

Technologies Used

- **Python**
 - **Pandas** – Data cleaning and manipulation
 - **NumPy** – Numerical operations
 - **Matplotlib & Seaborn** – Static visualizations
 - **Plotly Express** – Interactive visualizations
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Dataset

- Dataset: *Telco Customer Churn Dataset*
 - Contains customer details such as:
 - Demographics (gender, senior citizen)
 - Account information (tenure, contract type)
 - Services (internet service, payment method)
 - Billing (monthly & total charges)
 - Target variable: **Churn**
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Data Preprocessing

- Loaded dataset using Pandas
- Converted **TotalCharges** to numeric format
- Handled missing values by dropping null entries

- Removed unwanted columns (e.g., unnamed columns)
 - Checked:
 - Data types
 - Duplicates
 - Null values
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Interactive Visualizations

Scatter Plots (Bubble Charts)

1. **Monthly Charges vs Total Charges**
 - Bubble size represents tenure
 - Color indicates contract type
2. **Total Charges vs Tenure**
 - Bubble size represents monthly charges
 - Color indicates churn status

Insights:

- Customers with higher tenure tend to have higher total charges
 - Churn is more common among low-tenure customers
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Heatmap Visualization

- Created an interactive **heatmap** to visualize correlations between numerical features
- Helps in identifying strong positive or negative relationships between variables
- Provides a clearer understanding of how features like tenure, monthly charges, and total charges are related

Insights:

- Strong correlation observed between **tenure and total charges**
 - Moderate relationship between **monthly charges and total charges**
 - Helps in identifying key variables influencing churn patterns
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Treemap Visualizations

- Visualized hierarchical relationships:
 - Contract → Internet Service → Payment Method
- Included:
 - Total revenue contribution
 - Churn rate

Insights:

- Certain combinations of services and payment methods show higher churn rates
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Line Plot (Trend Analysis)

- Smoothed line plot of **Total Charges vs Tenure**
- Grouped by:
 - Contract type
 - Churn status

Insights:

- Long-term contracts show more stable revenue trends
 - Churn customers show different spending patterns over time
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Pie Charts

- Top Contract Types
- Payment Methods distribution

Insights:

- Majority of customers fall under specific contract types
 - Payment method preferences vary significantly
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Insights

- Customers with low tenure (new customers) have a significantly higher likelihood of churning, indicating that the early stage of the customer lifecycle is the most critical period where businesses need to focus on engagement, onboarding experience, and satisfaction.

- Customers using month-to-month contracts show the highest churn rates compared to those with one-year or two-year contracts, suggesting that lack of long-term commitment makes it easier for customers to leave the service.
 - Higher monthly charges are associated with increased churn risk, which implies that pricing plays an important role in customer retention and customers may switch to more affordable alternatives if they feel the cost is too high.
 - Customers with longer tenure tend to generate higher total revenue and are less likely to churn, showing that loyal customers are more valuable and stable for the business over time.
 - Different payment methods and internet service types show variation in churn behavior, indicating that customer preferences and service quality can influence their decision to stay or leave.
 - The relationship between Total Charges, Monthly Charges, and Tenure highlights that customers who stay longer gradually contribute more revenue, while churned customers often exit before reaching high total value.
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Conclusion

- This project demonstrates how visualizing customer data can reveal important patterns related to churn behavior. By analyzing factors such as tenure, monthly charges, contract type, and payment methods, it becomes evident that customers with shorter tenure and flexible contracts are more likely to leave. The use of multiple visualizations helped in clearly understanding these trends and relationships.
- Overall, the insights gained from this analysis can support businesses in making better decisions to reduce churn. By identifying high-risk customers early and improving services or pricing strategies, companies can enhance customer satisfaction and retention. This project shows the importance of data-driven approaches in solving real-world business problems.